Stroke

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Stroke is a major cause of death and disability globally. Diagnosis depends on clinical features and brain imaging to differentiate between ischaemic stroke and intracerebral haemorrhage. Non-contrast CT can exclude haemorrhage, but the addition of CT perfusion imaging and angiography allows a positive diagnosis of ischaemic stroke versus mimics and can identify a large vessel occlusion target for endovascular thrombectomy. Management of ischaemic stroke has greatly advanced, with rapid reperfusion by use of intravenous thrombolysis and endovascular thrombectomy shown to reduce disability. These therapies can now be applied in selected patients who present late to medical care if there is imaging evidence of salvageable brain tissue. Both haemostatic agents and surgical interventions are investigational for intracerebral haemorrhage. Prevention of recurrent stroke requires an understanding of the mechanism of stroke to target interventions, such as carotid endarterectomy, anticoagulation for atrial fibrillation, and patent foramen ovale closure. However, interventions such as lowering blood pressure, smoking cessation, and lifestyle optimisation are common to all stroke subtypes.

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Introduction

Stroke is a common disease, with one in four people affected over their lifetime, and is the second leading cause of death and third leading cause of disability in adults worldwide.[1] Substantial advances in therapy have occurred in the past 5 years, particularly for the acute treatment of ischaemic stroke. New strategies for preventing recurrence have also been identified. This Seminar outlines the diagnosis and management of ischaemic stroke and intracerebral haemorrhage in contemporary stroke units.

Definition of stroke

Stroke is defined as a neurological deficit attributed to an acute focal injury of the CNS (ie, brain, retina, or spinal cord) by a vascular cause.[2] Most strokes are ischaemic due to reduced blood flow, generally resulting from arterial occlusion. A rarer type of ischaemic stroke is venous infarction due to occlusion of cerebral veins or venous sinuses. The remaining 10–40% of stroke presentations, depending on regional epidemiology, are haemorrhagic and result from the rupture of cerebral arteries.[3,4] These haemorrhages can be intracerebral or subarachnoid; subarachnoid haemorrhages typically result from a ruptured aneurysm and are out of the scope of this Seminar. Ischaemic stroke is differentiated from transient ischaemic attack by the presence of an infarct on brain imaging. Patients diagnosed with transient ischaemic attack, by use of former clinical definitions that were based on symptom resolution within 24 h, have evidence of infarction on diffusion-weighted MRI in approximately 40% of cases and represent a group who are at high risk for recurrent stroke.[5]

Diagnosis of stroke and mimics

The key clinical feature of stroke is the sudden onset of a focal neurological deficit. The timing of this sudden onset can be masked if the patient awakens with stroke symptoms or if the onset is unwitnessed and the patient is unable to communicate or does not have the insight to recognise the timing of deficit. The time of stroke onset is therefore defined as the time that the patient was last known to be well.

Knowledge of neuroanatomical structures and vascular territories allows localisation and estimation of the size of the affected territory; patterns, such as right hemiparesis with aphasia due to occlusion of the left middle cerebral artery, are common and well recognised. Stroke symptoms that are under-recognised, such as nausea, vomiting, vertigo, and decreased level of consciousness, are more common in the setting of occlusions in the posterior circulation.[6]

Sudden onset of neurological deficits generally indicates a vascular cause, although seizures, specifically focal impaired awareness seizures or a postictal state, can also produce sudden onset of symptoms. Additionally, migraine with aura or hemiplegic migraine can also lead to sudden onset of focal neurological symptoms, but this should be a diagnosis of exclusion. Functional (psychogenic) deficits, such as conversion disorder, can mimic stroke. Occasionally, space-occupying lesions can present suddenly if they cause a seizure or bleeding. Other mimics, which would typically not show abrupt onset with an adequate patient history, include toxic metabolic derangements

Search strategy and selection criteria

We searched the Cochrane Library, MEDLINE, and Embase for articles published in English between Jan 1, 2015, and Dec 31, 2019. We used the search terms "ischaemic/ischemic stroke" or "intracerebral haemorrhage/hemorrhage", and "clinical trial" or "meta-analysis". We also searched the reference lists of articles identified by this search strategy and selected those that we judged to be relevant. We largely selected publications in the past 5 years but did not exclude commonly referenced and highly regarded publications that were older. Review articles are cited to provide readers with more details and references than this Seminar is able to. Our reference list was modified on the basis of comments from peer reviewers.

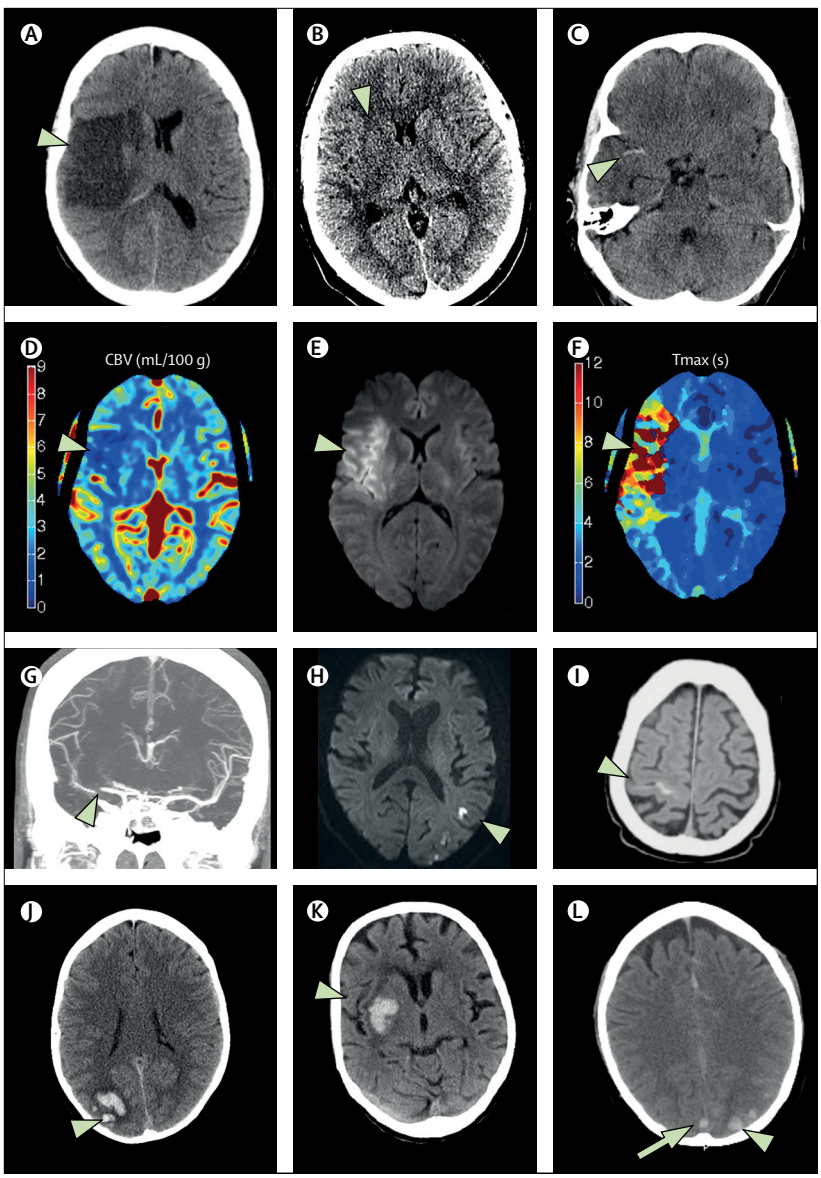

Figure 1: **Stroke diagnosis using neuroimaging**

(A) Ischaemic stroke established in the territory of the right middle cerebral artery 12 h after onset (arrowhead). A patient with left hemiparesis onset 2 h before scan (B–G) showing subtle loss of differentiation between grey and white matter (arrowhead) in the basal ganglia (B) and hyperdense thrombus (arrowhead) in the right middle cerebral artery (C). CT perfusion showing reduced CBV (arrowhead) in the right insular region (D) corresponding to region of diffusion restriction (arrowhead; most likely irreversibly injured) on MRI (E). CT perfusion showing delayed Tmax (arrowhead; substantially delayed Tmax [ie, >6 s] indicates brain tissue that is critically hypoperfused, functionally impaired, and potentially at risk of infarction in the absence of reperfusion) in the right middle cerebral artery territory (F) corresponding to intracranial occlusion of the right middle cerebral artery (arrowhead) on CT angiogram (G). (H) Diffusion MRI lesions (arrowhead) in a patient with two 5 min episodes of aphasia that fully resolved—now defined as ischaemic stroke rather than transient ischaemic attack. (I) Focal subarachnoid haemorrhage (arrowhead) related to amyloid angiopathy presenting as transient parasthesias on the left side (differential diagnosis of transient ischaemic attack). Lobar intracerebral haemorrhage (arrowhead) in a patient with amyloid angiopathy (J) and intracerebral haemorrhage (arrowhead) in the right basal ganglia most likely resulting from deep perforating vasculopathy (K). (L) Thrombosis in the cerebral venous sinus with hyperdense sagittal sinus (arrow) and haemorrhagic venous infarction (arrowhead). CBV=cerebral blood volume. Tmax=time to maximum.

(eg, hypoglycaemia). Particularly in patients with a previous history of stroke, the previous neurological deficit can return with intercurrent illness.[7]

Brain imaging (CT scan or MRI) crucially complements the clinical examination to differentiate the stroke subtype and mechanism. Clinical symptoms and signs alone cannot reliably differentiate ischaemic stroke from intracerebral haemorrhage, and management sharply diverges between these two conditions. In ischaemic stroke, the presence of a large vessel occlusion, defined as occlusion in the internal carotid artery, proximal middle cerebral artery, or basilar artery, determines the most appropriate reperfusion therapy. The secondary prevention of large artery atherosclerotic disease versus cardioembolic stroke also differs substantially. Imaging of the brain and its vascular tree is therefore one of the most urgent priorities when patients present to hospital with suspected stroke (figure 1).

Epidemiology and risk factors

The 2016 Global Burden of Disease data that were published in 2019 indicate that one in four people will have a stroke in their lifetime.[3] There are estimated to be 9·6 million ischaemic strokes and 4·1 million haemorrhagic strokes (including intracerebral and subarachnoid haemorrhage) globally each year, with a relatively stable incidence adjusted for age in high-income countries but an increasing incidence in low-income and middle-income countries.[3] The absolute incidence is expected to increase with an ageing population.

Estimations indicate that approximately 90% of strokes are attributable to modifiable risk factors.[8] Stroke shares many risk factors with other cardiovascular diseases although their relative importance varies. The most potent risk factor for stroke is high blood pressure, which applies to both ischaemic stroke and intracerebral haemorrhage. Smoking, diabetes, hyperlipidaemia, and physical inactivity are also significant risks and require interventions that are regulatory and are based in the community to alter lifestyle and the environment, as well as individual treatment.[9] Atrial fibrillation, a specific risk factor for ischaemic stroke, is increasing in detection and prevalence.[10] Strokes related to atrial fibrillation tend to be larger and more disabling than are strokes due to other mechanisms.[11]

Pathophysiology

Ischaemic stroke

Most ischaemic stroke is due to embolism, either from atherosclerotic plaque in the aortic arch or in the cervical arteries or from the heart (panel, figure 2). Intracranial atherosclerosis with in-situ thrombosis is also an important mechanism of stroke, particularly in Asian and Black ethnic groups.[12] Small vessel disease causes small subcortical infarcts (ie, lacunar stroke) and deep intracerebral haemorrhage. Cervical artery dissection is one of the common causes of stroke in younger patients (eg, <60 years), and arterial

Panel: Major causes of stroke

Atherosclerosis:
- Aortic arch or cervical arteries
- Intracranial arteries

Cardioembolism:
- Atrial fibrillation
- Akinetic myocardial segment
- Patent foramen ovale
- Endocarditis

Small vessel disease

Other causes:
- Other arterial diseases (eg, dissection, vasculitis)
- Haematological diseases (eg, antiphospholipid syndrome, polycythaemia rubra vera, essential thrombocytosis)

inflammation can also cause stroke (eg, inflammatory arteriopathy after infection is a major cause of paediatric stroke and can also occur after herpes zoster in adults).

When a cerebral artery is occluded and blood flow decreases below a critical level, neuronal electrical function ceases and a clinical deficit develops.[13] If cerebral blood flow is severely reduced, then irreversible tissue injury will ensue rapidly. However, in many patients, collateral blood supply via leptomeningeal anastomoses or the circle of Willis can be sufficient to maintain cellular viability for a period of time. These hibernating, but potentially salvageable, brain regions are termed the ischaemic penumbra. Reperfusion therapies restore blood flow to the ischaemic penumbra and substantially reduce disability after ischaemic stroke. The salvageable ischaemic penumbra can be identified non-invasively by use of the mismatch between the ischaemic core, which is irreversibly injured (estimated using diffusion MRI or severely reduced cerebral blood flow on CT perfusion), and the critically hypoperfused region (estimated as the region of substantially delayed blood flow arrival).[14,15] This estimation of salvageable tissue by use of perfusion imaging has been successfully used to identify patients who would benefit from reperfusion therapies beyond the standard time windows of 6·0 h for endovascular thrombectomy and 4·5 h for intravenous thrombolysis.[16,17]

Ischaemia and reperfusion can cause secondary injury. Although reperfusion injury is well described in animal models, it has been less easily recognised in humans since the benefits of reperfusion usually far outweigh the detrimental effects.[18] However, symptomatic haemorrhagic transformation and malignant oedema are two clinical manifestations of reperfusion injury. Glutamate excitotoxicity, free radical injury, and matrix metalloprotease degradation of the blood–brain barrier are just some of the described mechanisms of secondary injury after reperfusion.[19]

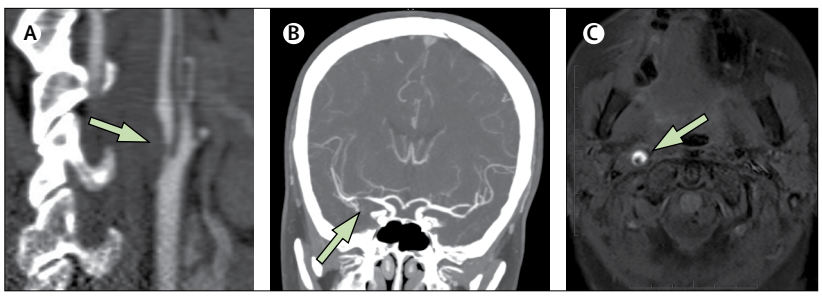

Figure 2: **Determining stroke mechanism**
(A) CT angiography showing atherosclerosis of the internal carotid artery. (B) Intracranial atherosclerotic disease. (C) Fat saturated T1 MRI showing intramural hyperintensity diagnostic of carotid artery dissection.

Intracerebral haemorrhage

The most common cause of intracerebral haemorrhage is deep perforating vasculopathy related to high blood pressure,[20] with cerebral microbleeds (seen on MRI) and clinical haemorrhages most often affecting the basal ganglia, cerebellum, pons, or thalamus. Another major cause is amyloid angiopathy; these haemorrhages are typically lobar and occur in older patients (ie, aged >55 years but most frequently in patients aged 70–80 years) with MRI evidence of cortical microbleeds and superficial haemosiderosis. Vascular malformations (eg, arteriovenous malformations, cavernous malformations, and dural arteriovenous fistulae) and mass lesions (eg, metastatic tumours) should be ruled out with neuroimaging, especially in younger patients (eg, <60 years) or without evidence of vasculopathy on MRI.

The detrimental effects of intracerebral haemorrhage due to mass effect from the haematoma itself are readily recognised. The oedema that subsequently develops (and can increase for up to 2 weeks) also contributes to injury from mass effect, and toxicity from thrombin and iron are thought to be key contributors to the development of oedema.[20]

Acute management

Acute management of patients with stroke should occur in a stroke unit that is organised and geographically defined. Care in a stroke unit has been clearly shown to increase survival without disability for patients of all ages, severities, and stroke subtypes,[21] and comprises an expert integrated medical, nursing, and allied health team applying evidence-based clinical protocols (table). Care in a stroke unit is the foundation on which acute stroke interventions can be delivered. The aims are to reduce complications, such as aspiration pneumonia, venous thromboembolism, and pressure sores; commence early rehabilitation; and institute targeted secondary prevention. Protocolised nursing management of fever, glucose, and swallowing reduced mortality in one cluster-randomised trial.[32] However, a 2019 trial did not show any benefit of more intensive glucose control versus standard management after stroke.[33]

	Treatment patients with outcome, %	Control patients with outcome, %	Odds ratio (95% CI)	Absolute difference, %
Care in a stroke unit				
Death or dependency (mRS 3–6)[21]	52·4%	60·9%	0·75 (0·66–0·85)	8·5%
Ischaemic stroke				
Alteplase thrombolysis, non-contrast CT brain selection[22,23]				
mRS 0–1 in patients treated 0·0–3·0 h after stroke onset	32·9%	23·1%	1·75 (1·35–2·27)	9·8%
mRS 0–1 in patients treated 3·0–4·5 h after stroke onset	35·3%	30·1%	1·26 (1·05–1·51)	5·2%
SICH	3·7%	0·6%	6·67 (4·11–10·84)	3·1%
Fatal SICH	2·7%	0·4%	7·14 (3·98–12·79)	2·3%
Mortality in patients treated 0·0–3·0 h after stroke onset	22·2%	21·8%	1·00 (0·81–1·24)	0·4% (p=0·70)
Mortality in patients treated 3·0–4·5 h after stroke onset	16·9%	15·9%	1·14 (0·95–1·36)	1·0% (p=0.96)
Alteplase thrombolysis >4·5 h after stroke onset in patients selected by use of perfusion imaging[17]				
mRS 0–1	36·2%	25·8%	2·06 (1·17–3·62)	10·4%
mRS 0–2	50·7%	39·7%	2·22 (1·25–3·94)	11·0%
SICH	4·6%	0·7%	7·29 (0·88–60·18)	3·9% (p=0·067)
Mortality	13·2%	10·5%	1·28 (0·60–2·73)	2·7% (p=0·52)
Endovascular thrombectomy initiated 0·0-6·0 h after stroke onset[24]				
mRS 0–1	26·9%	12·9%	2·72 (1·99–3·71)	14·0%
mRS 0–2	46·0%	26·5%	2·71 (2·07–3·55)	19·5%
SICH	4·4%	4·3%	1·07 (0·62–1·84)	0·1% (p=0·81)
Mortality	15·3%	18·9%	0·73 (0·47–1·13)	3·6% (p=0·16)
Endovascular thrombectomy initiated 6·0–24·0 h after stroke onset in patients selected by the use of perfusion imaging[25]				
mRS 0–2	46·7%	14·8%	5·01 (3·07–8·17)	31·9%
SICH	6·0%	3·7%	1·67 (0·64–4·35)	2·3% (p=0·29)
Mortality	16·6%	21·7%	0·71 (0·34–1·51)	5·1% (p=0·38)
Hemicraniectomy[26]				
mRS 4–6	56·9%	78·6%	0·33 (0·13–0·86)	21·7%
Mortality	21·6%	71·4%	0·10 (0·04–0·27)	49·8%
Aspirin administered <48·0 h after stroke onset[27]				
mRS 0–2	54·4%	53·1%	1·05 (1·01–1·10)	1·3%
Intracerebral haemorrhage				
Intensive blood pressure lowering[28]				
mRS 3–6	52·0%	55·6%	0·87 (0·75–1·01)	3·6% (p=0·059)
Surgical evacuation overall[29]				
Death or disability	59·4%	67·4%	0·72 (0·61–0·84)	8·0%
Mortality	27·3%	31·8%	0·82 (0·69–0·97)	4·5%
Surgery commenced within 0·0–8·0 h of stroke onset[29]				
Death or disability*	70·3%	79·2%	0·59 (0·42–0·84)	8·9%
Minimally invasive surgery[30]				
Death or disability†	47·4%	65·4%	0·59 (0·42–0·84)	18·0%

SICH is defined as parenchymal haematoma occupying >30% of the infarcted territory with substantial mass effect combined with an increase of ≥4 points in National Institutes of Health Stroke Scale score, as used in the Safe Implementation of Thrombolysis in Stroke-Monitoring Study.[31] mRS=modified Rankin scale. SICH=symptomatic intracerebral haemorrhage. *Component studies used different outcomes: composite of death, vegetative or severe disability outcome on Glasgow Outcomes Score; mRS ≥3; or Barthel Index ≤90 in the 0·0–8·0 h analysis. †Component studies used different outcomes: composite of mRS ≥3; or Barthel Index ≤60.

Table: **Patient outcomes following acute interventions for stroke**

Acute treatments for ischaemic stroke

Intravenous thrombolysis with recombinant human tissue plasminogen activator (alteplase) aims to reperfuse the ischaemic brain by converting plasminogen (PLG) to plasmin, which can dissolve the thrombus that is causing the stroke. Alteplase was first shown to reduce disability in the NINDS part A and B trials[34] when administered within 3·0 h of stroke onset. The treatment window was subsequently extended to 4·5 h, although the benefit reduces rapidly with increasing time after stroke onset (table).[22] When alteplase is delivered within 3·0 h of onset, approximately one in four patients have reduced disability, which decreases to one in seven patients between 3·0 h and 4·5 h.[35] This benefit includes the effect of the approximately 2% absolute risk of fatal intracerebral haemorrhage. A large meta-analysis of individual patient data established that, although age and clinical severity measured by use of the National Institutes of Health Stroke Scale are strongly prognostic, the treatment benefit of alteplase is preserved across the spectrum of these variables.[22] Patients with mild but disabling symptoms benefit from thrombolysis. However, the prematurely terminated 2018 PRISMS trial[36] showed no evidence of benefit in patients with symptoms that were judged to be non-disabling at presentation, and who were selected on the basis of non-contrast CT brain imaging and clinical characteristics. Trials selecting patients with non-disabling symptoms but with arterial occlusion or perfusion abnormality are ongoing (eg, TEMPO-2, NCT02398656).

In 2019, the use of CT or perfusion MRI was established to select patients between 4·5 h and 9·0 h from the time that they were last seen to be well (or within 9·0 h of the midpoint of sleep if they awoke with stroke) if they had imaging evidence of salvageable brain tissue.[17,37] This subset of patients derives at least as much benefit with similar risk of fatal intracerebral haemorrhage to those treated 0·0–3·0 h after stroke onset, and the ability to select patients using CT-based imaging puts this treatment approach within reach of most hospitals that are capable of thrombolysis intervention. Thrombolysis also improved outcomes in patients with unknown time of stroke onset (including those waking up with stroke) in whom MRI showed diffusion lesions that were not yet hyperintense on fluid-attenuated inversion recovery (FLAIR).[38] This diffusion-FLAIR mismatch indicates that the patient is likely to be within 4·5 h of stroke onset. Compared with CT perfusion, MRI diffusion-FLAIR mismatch is more likely to detect patients with lacunar stroke, and who could benefit from thrombolysis.[39] However, the requirement for urgent MRI reduces the applicability of the MRI diffusion-FLAIR technique in many regions.

The 0·9 mg/kg licensed dose of alteplase in most regions was based on data from small studies, and the licensed dose in Japan is 0·6 mg/kg.[40] A randomised trial comparing these two doses did not show non-inferiority

of 0·6 mg/kg, although symptomatic intracerebral haemorrhage was reduced from 2% to 1%.[41] There are plans to explore the lower dose of 0·6 mg/kg further in patients who are felt to be at high risk of bleeding (eg, concurrent antiplatelet use). However, as of June, 2020, all guidelines outside of those in Japan recommend 0·9 mg/kg alteplase (maximum 90·0 mg) delivered as a 10% bolus followed by 90% infused over 1 h.

Tenecteplase is a genetically modified form of alteplase that has a longer half-life, permitting a single bolus administration (rather than bolus and 1 h infusion of alteplase) and greater fibrin specificity and resistance to plasminogen activator inhibitors than does alteplase.[42] Randomised trial data suggest that tenecteplase is at least as safe and as effective in patients with stroke[43] and that patients with large vessel occlusion had higher rates of reperfusion with tenecteplase versus with alteplase.[44,45] Although 0·25 mg/kg and 0·40 mg/kg have been used in trials, the EXTEND-IA TNK part 2 trial[46] indicated that there was no advantage in increasing the dose from 0·25 mg/kg to 0·40 mg/kg. Because many patients now require transfer between hospitals, single bolus administration simplifies the transport process and ensures that the full dose of thrombolytic agent is given, since alteplase infusions could be interrupted in transit. Tenecteplase use also avoids the common gap between administration of the bolus and infusion of alteplase, which, given the short half-life of alteplase, can mean that the desired plasma concentration is not sustained. Although tenecteplase has entered guidelines in Europe,[47] the USA,[48] and Australia[49] as a possible alternative to alteplase, it is not currently licensed for use as a stroke treatment outside of India, where a generic form of tenecteplase was licensed on the basis of non-randomised data,[50] and its biosimilarity is debated.[51] Ongoing phase 3 trials of tenecteplase aim to definitively establish the role of tenecteplase for patients with stroke (TASTE, ACTRN12613000243718; ATTEST2, NCT02814409; ACT, NCT03889249), including potentially combining tenecteplase with endovascular thrombectomy in patients presenting later than 4·5 h after stroke onset (TIMELESS, NCT03785678; ETERNAL, NCT04454788).

Intravenous thrombolysis is the most accessible reperfusion therapy for stroke because endovascular thrombectomy is restricted to major stroke centres and completely unavailable in many parts of the world. Several trials are testing whether thrombolysis can be safely omitted for patients with large vessel occlusion who present directly to a centre that is capable of thrombectomy. The first trial to report results showed similar outcomes between groups that narrowly met a generous 20% non-inferiority margin (DIRECT-MT)[52] and further studies are ongoing (SWIFT DIRECT, NCT03192332; DIRECT SAFE, NCT03494920; MR CLEAN-NO IV, ISRCTN80619088). The current standard of care is to give thrombolysis and proceed to thrombectomy as quickly as possible. Therefore, approaches to improve the effectiveness of intravenous thrombolysis have great potential value. Current trials are exploring the addition of adjuvant agents (eg, eptifibatide or argatroban in the MOST trial, NCT03735979). Novel drugs that dissolve clots and target other structural components of thrombi (eg, von Willebrand factor and neutrophil extracellular traps), inhibitors of fibrinolysis (eg, CPB2 and α2-AP), and mechanical strategies, such as sonothrombolysis, are also under investigation.[53]

Endovascular thrombectomy is another type of reperfusion therapy. After several trials did not show any benefit with endovascular thrombectomy in 2013,[54–56] five trials published in 2015 established endovascular thrombectomy as one of the most powerful treatments to reduce disability in any specialty of medicine (table).[57–61] The benefits shown by these five trials were driven by a combination of improved devices (which allowed faster, more effective reperfusion), improved patient selection (requiring at least a documented large vessel occlusion on non-invasive angiography), and faster treatment workflow. A meta-analysis of individual patient data emphasised the importance of time, with one in 100 patients worse off for every 4 min delay in reperfusion after arriving in the emergency department.[62]

In what might superficially appear to contradict this crucial relationship between time to treatment and functional outcome, trials in 2018 established the benefits of endovascular thrombectomy up to 24 h after the time that the patient was last known to be well, if perfusion imaging was favourable.[16,63] The key point is that the proportion of patients who have favourable perfusion imaging decreases over time, and so the urgency to evaluate and treat rapidly still exists; fast treatment will maximise the proportion of patients who have salvageable brain tissue. However, if a patient is unavoidably delayed in presenting to hospital and they still have favourable imaging, they will derive at least as much treatment benefit as patients who receive treatment within 0–6 h of stroke onset (table).

Endovascular thrombectomy, analogous to thrombolysis, is of generalised benefit across the spectrum of age and clinical severity.[24] The benefit of thrombectomy is uncertain in patients who are mildly affected (only 14 patients with National Institutes of Health Stroke Scale score <6 were enrolled in completed trials), and ongoing trials are addressing the use of endovascular thrombectomy in this population (eg, ENDO-LOW, NCT04167527; MOSTE, NCT03796468). Patients with occlusions in the internal carotid artery and proximal middle cerebral artery (M1 segment; figure 3) benefit from endovascular thrombectomy, including those with tandem cervical internal carotid artery and intracranial occlusion. Once the middle cerebral artery has bifurcated (M2 segments), the benefit is potentially reduced because a smaller territory is at risk and there is an increased effect of thrombolysis. Additionally, the risk of arterial injury might be increased because of increased technical difficulty of thrombectomy in smaller, more tortuous

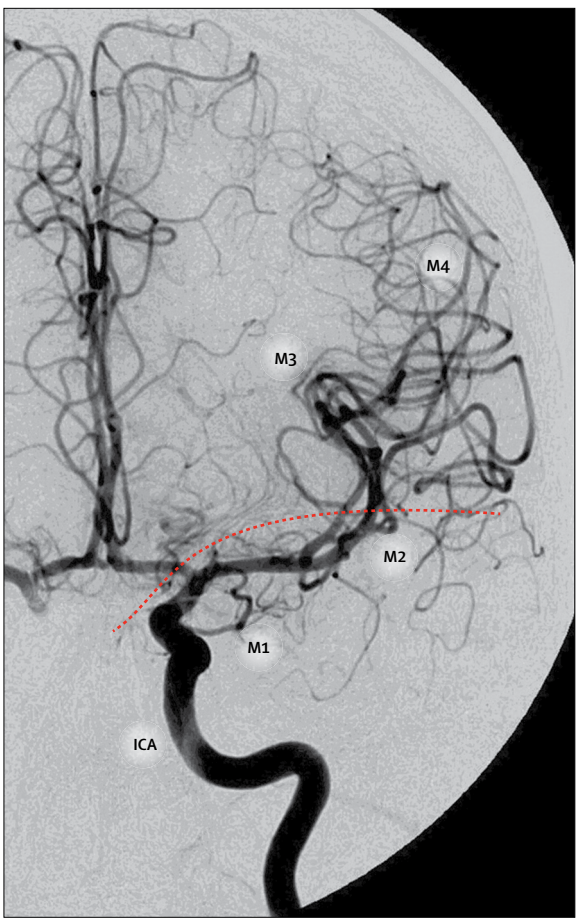

Figure 3: **Intracranial vasculature**
The evidence supports endovascular thrombectomy in the internal carotid artery, M1 segment of the middle cerebral artery, and selected patients with proximal M2 segment occlusion (approximated by the dotted red line). Distal vessels could become more accessible with technological developments. ICA=internal carotid artery.

vessels. Data indicate the benefit of thrombectomy in patients with occlusions in the large, more proximal M2 branches, who have clinically significant neurological deficits, but treatment decisions need to be individualised for these patients.[26] Technology continues to evolve and thrombectomy in more distal vessels will consequently require further evaluation.

Thrombectomy in the basilar artery is recommended by guidelines[47–49] but convincing randomised data are scarce. The Chinese BEST trial[64] reported a benefit of approximately 20% in an as-treated analysis but this result was confounded by a high crossover rate from control to intervention. The BASICS trial[65] has been reported in an abstract and overall results were neutral. The more severely affected subgroup of patients (National Institutes of Health Stroke Scale score ≥10) did appear to benefit from thrombectomy. A second Chinese trial (BAOCHE, NCT02737189) is due to be completed soon.

Whether endovascular thrombectomy benefits patients with large areas of irreversibly injured brain (ischaemic core) is uncertain. The ischaemic core can be estimated (in order of increasing precision) using hypodensity on non-contrast CT (loss of the normal differentiation between grey and white matter), severely reduced blood flow on CT perfusion, or diffusion restriction on MRI. Increasing ischaemic core volume is undoubtedly associated with a worse prognosis. However, a crucial question is whether a meaningful treatment benefit from thrombectomy exists in patients with a large ischaemic core volume (eg, >70 mL or >100 mL). Data indicate that functional improvement is noted for at least a proportion of patients with a large ischaemic core,[66,67] although few of these patients were included in the pivotal trials. Several ongoing trials are attempting to address this issue (eg, TENSION, NCT03094715; SELECT-2, NCT03876457; TESLA, NCT03805308; LASTE, NCT03811769). One challenge is that futile treatment has sometimes been defined as not enabling a return to independence. This definition ignores clinically and economically meaningful shifts from death or disability that requires residence in a nursing home to requiring a moderate level of assistance that is compatible with living at home.

Notably, even after successful endovascular thrombectomy, approximately half of patients with large vessel occlusion do not regain independent function.[24] This issue has spurred a new generation of trials in neuroprotection and recovery enhancement. Before trials showing the benefit of thrombectomy, many neuroprotection trials did not translate the seemingly potent effects in animal models to humans, which created intense scepticism about neuroprotection in humans. The Stroke Therapy Academic Industry Roundtable (STAIR) criteria were created in 1999 in response to preclinical studies that did not show translation to humans and were updated in 2009 with the aim of introducing greater rigour into preclinical research.[68] One of the first completed phase 3 trials that followed the STAIR pathway, and also capitalised on the new era of endovascular thrombectomy, was the ESCAPE NA-1 trial[69] of nerinetide, a PSD95 (DLG4) inhibitor that aims to reduce glutamate excitotoxicity. Nerinetide was tested in multiple preclinical models, including primates, using randomisation and blinding.[70] A subsequent phase 2 clinical trial[71] showed that nerinetide significantly reduced incidental diffusion lesions in patients undergoing endovascular aneurysm repair. The phase 3 trial[69] enrolled 1105 patients and, although not statistically significant overall, the results suggested reduced disability in patients who had not also received alteplase. Alteplase treatment was associated with substantially lower serum concentrations of nerinetide than in patients who did not receive alteplase, due to protease activation.

Another example of adjuvant therapy is the treatment of malignant oedema in patients with large hemispheric infarction. Hemicraniectomy reduces death and disability in these patients, mostly younger than 60 years, who are at risk of transtentorial herniation leading to brainstem compression due to large infarcts in the

middle cerebral artery territory.[72] Intravenous glibenclamide inhibits SUR-1 (ABCC8) and a phase 3 trial (CHARM, NCT02864953) is underway to test this pharmacological approach to oedema. In patients with large cerebellar infarcts, posterior fossa craniectomy is a life-saving procedure to decompress the brainstem and the fourth ventricle.[48]

Acute treatments for intracerebral haemorrhage

Other than care in a stroke unit, intensive lowering of blood pressure at an early stage to approximately 140 mm Hg systolic is the only evidence-based treatment for intracerebral haemorrhage.[28] Even then, the extent of absolute reduction in disability was 3·6% and the primary outcome of the trial was not significant (table). Lowering the blood pressure more intensively to 120 mm Hg was not beneficial and led to increased renal adverse events.[73] Although pooled as-treated analysis showed improved outcomes with a reduction to 120 mm Hg, this result could have been confounded by incomplete adjustment of prognostic variables.[74]

Reversal of antithrombotic medications is another acute treatment for intracerebral haemorrhage. The effects of warfarin can be reversed with prothrombin factor concentrate and vitamin K. Unfractionated heparin can be reversed with protamine. Dabigatran can be reversed almost instantaneously with idarucizumab, and low molecular weight heparin and the anti-Xa direct oral anticoagulants apixaban and rivaroxaban can be reversed using andexanet alfa. Some data indicate that faster normalisation of coagulation status by use of prothombin factor concentrate, rather than fresh frozen plasma, in patients treated with warfarin is associated with less haematoma expansion and improved outcomes.[75] However, platelet transfusion for patients taking antiplatelet agents and not undergoing surgery worsened outcomes, which is thought to be related to immune activation.[76]

Haemostatic therapies have also been trialled for acute treatment of intracerebral haemorrhage. Trials of tranexamic acid[77] and recombinant activated factor VII[78,79] in intracerebral haemorrhage patients with normal coagulation have not shown a significant effect. Further trials of earlier treatment with these agents are ongoing (eg, STOP-MSU, NCT03385928; FASTEST, NCT03496883). Tranexamic acid, an inexpensive drug, has shown encouraging results in traumatic intracerebral haemorrhage, reducing mortality in patients treated within 3 h of injury.[80]

Surgical interventions are another option for acute treatment. Surgical evacuation of the haematoma has been assessed in multiple trials, which were often confounded by high crossover rates from control to intervention. Heterogeneous results have prevented mainstream adoption of surgical treatment, although a meta-analysis[29] suggests an overall benefit (table). In some countries, such as Japan, minimally invasive surgery is routine and a meta-analysis[30] has suggested promising results. The MISTIE III trial[81] involved inserting a catheter into the haematoma after showing stable volume on serial CT scans (median 46 h after onset) and instilling alteplase. This treatment reduced the volume of the haematoma over approximately 4 days. Overall, the trial did not show a significant effect, but the subgroup with successful haematoma removal to less than 15 mL residual volume did have better functional outcomes than those of standard care. MISTIE III offers hope that surgical evacuation techniques that are consistently effective might translate to improved patient outcomes, and several trials are underway (ENRICH, NCT02880878; MIND, NCT03342664; EVACUATE, NCT04434807). Hemicraniectomy is also being explored for intracerebral haemorrhage (SWITCH, NCT02258919).

Acute systems of care

Faster treatment would deliver the greatest benefit from reperfusion therapies.[22,62] Implementing fast treatment requires system engineering across the prehospital and emergency department continuum of care, with the prehospital setting comprising the largest component of time delay between stroke onset and reperfusion.[62]

Increasing community recognition of stroke reduces the time taken for a patient to present to medical care. The Face, Arm, Speech, Time to act (known as the FAST mnemonic) message is used internationally to teach the general public about the signs of stroke and emphasise the need to call an ambulance immediately. Approximately 89% of patients with stroke will have face, arm, or speech affected.[82]

The aim of prehospital care by paramedics is to recognise stroke with high sensitivity and rapidly transport the patient to an appropriate hospital that is equipped to deal with stroke. Paramedics should give prenotification to the receiving emergency department to allow the stroke team to meet the patient at the door and proceed directly to CT scan.[83] Ideally, paramedics would also use severity-based triage tools[84–86] to identify suspected large vessel occlusion and transport those patients directly to a centre capable of endovascular thrombectomy, provided that the travel time is not excessive (eg, is less than 30 min) and the additional time taken will not disqualify the patient from thrombolysis.[87] These triage tools combine various elements of face, arm, speech, and hemispatial inattention examination findings and detect most patients with large vessel occlusion. However, a proportion of patients with intracerebral haemorrhage and some patients with ischaemic stroke who require only thrombolysis would also be taken to a centre capable of endovascular thrombectomy, rather than their nearest stroke centre. Modelling has indicated that in most metropolitan geographies, bypass to a hospital capable of endovascular thrombectomy should deliver net benefit.[88] Randomised trials are ongoing to test this concept (RACECAT, NCT02795962; TRIAGE, NCT03542188).

The mobile stroke unit, an ambulance equipped with a CT scanner and personnel with stroke expertise, is another prehospital innovation that aims to reduce delays in treatment.[89] Approximately 30 units are currently operating worldwide, predominantly in metropolitan environments with high availability of resources. The ability to exclude intracerebral haemorrhage, commence thrombolysis in the field, and accurately triage patients with large vessel occlusion to hospitals capable of endovascular thrombectomy saves considerable time compared with assessment in the emergency department.[90] The B_PROUD part 1 study based in Berlin (NCT02869386) has been reported in an abstract and showed improved functional outcomes in patients treated on the mobile stoke unit.[91] B_PROUD part 2 (NCT03931616) and BEST-MSU (NCT02190500) are underway, aiming to show definitive improvement in clinical outcomes and cost savings.

Prenotification from paramedics needs to be passed on to the emergency department and the stroke team. Transporting the patient directly to a CT scanner on the ambulance stretcher prioritises the rate-limiting step in decision making and saves approximately 30 min in most studies compared with offloading the patient in an emergency department cubicle and then organising the CT scan.[92] However, an ongoing controversy in stroke care is the optimal imaging approach for fast but accurate treatment. A non-contrast CT brain image is all that is required for thrombolysis within 4·5 h and occlusion on CT angiogram is all that is required for thrombectomy within 6·0 h.[48] Reperfusion therapies should not be delayed for the sake of additional imaging. However, when treatment decisions are complicated by diagnostic uncertainty, mild deficits, or patient comorbidities, there can be diagnostic and prognostic advantages to gaining additional information from CT perfusion imaging, even within 6·0 h (figure 1).[66,93] Beyond 4·5 h, selection of patients who might benefit from thrombolysis requires data from CT perfusion, MR perfusion-diffusion, or MR diffusion-FLAIR imaging. Beyond 6·0 h, all trials establishing the benefit of thrombectomy have required perfusion data to identify patients who would benefit from delayed reperfusion.

Transfers between hospitals also need to be optimised. Globally, most patients who receive endovascular thrombectomy have been transferred after initially presenting to a hospital that does not offer endovascular thrombectomy. These patients have worse outcomes than those who present directly to a hospital capable of endovascular thrombectomy, largely because of delays in reperfusion.[94] Holding the original paramedic crew until after the CT scan has been done, to establish whether a secondary transfer is required, and streamlining referral pathways to a stroke centre capable of endovascular thrombectomy can reduce the time spent at the initial hospital (door-in door-out time).[95]

Secondary prevention

Ischaemic stroke and transient ischaemic attack

The general principles of secondary stroke prevention involve an approach to absolute cardiovascular risk with treatment of all risk factors in a patient who is, as a result of having had a stroke, at high risk of recurrent stroke and cardiovascular disease. However, secondary prevention also needs to be tailored to the specific mechanism of the incident stroke, and this requires thorough investigation for causative factors.

CT angiography from aortic arch to cerebral vertex is the favoured modality to assess atherosclerotic burden, cervical arterial dissection, and other arteriopathies. CT venography is required if there is suspicion of venous sinus thrombosis. Electrocardiogram (ECG) monitoring is required to detect atrial fibrillation, which is often paroxysmal and therefore difficult to capture.

The traditional approach of Holter monitoring for 24 h is inadequate and monitoring for a longer term significantly increases the diagnostic yield.[96] Loop recorders can be implanted to continuously monitor heart rhythm for 3 years, and simulation studies suggest that most atrial fibrillation that is detected occurs beyond the first month in which monitoring with non-invasive ECG might be applied.[97]

An ECG can provide clues to atrial fibrillation (eg, left atrial enlargement) and abnormalities might suggest akinetic left ventricular segments that pose a risk for mural thrombus. In patients younger than 60 years with no other identified cause of stroke, patent foramen ovale is now an accepted and treatable cause of stroke. Percutaneous closure of the patent foramen ovale has been shown to reduce recurrent stroke risk by approximately 1% per annum.[98] This risk appears to be cumulative year after year, leading to a significant reduction in absolute risk for young patients. A transthoracic echocardiogram or transcranial Doppler ultrasound with intravenous injection of agitated saline and Valsalva manoeuvre has high sensitivity to detect intracardiac (or intrapulmonary) shunting.[99] Transoesophageal echocardiography is then indicated to confirm the anatomical abnormality and plan closure.

Lowering blood pressure is crucial and epidemiological studies suggest that there is no lower limit to the benefit.[100] A reduction of approximately 9 mm Hg in systolic blood pressure was associated with a 23% (95% CI 10–35) relative reduction in ischaemic stroke risk.[101] Although targeting systolic blood pressure of less than 120 mm Hg versus less than 140 mm Hg reduced the risk of stroke in one trial, patients with a history of stroke were excluded.[102] More intensive lowering of blood pressure to less than 130 mm Hg systolic in patients with small subcortical strokes showed that recurrent stroke might be reduced[103] but further trial data are awaited that are specific to stroke. The optimal timing to lower blood pressure is undefined. Starting medication within 30 h of stroke did not improve outcomes[104] but commencing

medication before discharge is advisable to improve adherence and outcomes.[105] The amount that the blood pressure is lowered by appears to be more important than the class of medication used, although β blockers are not recommended as first-line medication[106] and can increase blood pressure variability, which is associated with increased risk of stroke.[107] Weight loss, physical activity, decreased dietary sodium intake, a diet rich in fruits, vegetables, and low-fat dairy, and low alcohol consumption are also recommended.[108]

High dose, high potency statins are indicated for most patients with ischaemic stroke.[109] This recommendation particularly relates to atherosclerotic mechanisms, although patients with atrial fibrillation might also benefit from statins.[110] A target of less than 1·8 mmol/L versus 2·3–2·8 mmol/L reduced the number of subsequent cardiovascular events.[111] In patients who are intolerant of statins, ezetimibe can be used, although data for cardiovascular outcome are weaker. PCSK9 inhibitors have strong evidence from trial data and are starting to be used in clinical practice but are expensive and require subcutaneous injection.[112]

Antiplatelet agents are indicated after ischaemic stroke unless there is atrial fibrillation, in which case anticoagulation is required. Trials of anticoagulation in patients with an embolic stroke caused by an uncertain source did not show a significant effect.[113,114] However, there is ongoing interest in whether atrial cardiopathy might be a risk factor for stroke in the absence of atrial fibrillation, and whether the atrial fibrillation could be an epiphenomenon (ARCADIA, NCT03192215).[115]

A combination of aspirin and clopidogrel in the short term, commenced with loading doses within 24 h and continued for 3 weeks, has been shown to reduce recurrent stroke after minor stroke and high risk transient ischaemic attack.[116,117] Dual antiplatelet therapy over a longer term increased the risk of bleeding without a significant benefit in stroke prevention. Aspirin is still an acceptable first-line agent, with clopidogrel[118] or aspirin–dipyridamole[119] being slightly more effective. There is ongoing interest in ticagrelor, particularly given pharmacogenomic variation among patients in the activation of clopidogrel to its active form.

For patients with non-valvular atrial fibrillation (ie, no mechanical prosthetic valve or moderate to severe mitral stenosis) and adequate renal function, the direct oral anticoagulants are generally preferred over warfarin because of convenience and the reduced risk of intracerebral haemorrhage.[120] Anticoagulation remains underused, leading to many preventable strokes. Perceived risk of bleeding might be overestimated, for example, in patients who have experienced falls. Many risk factors for bleeding are also risk factors for ischaemic stroke and so the risks tend to run in parallel with preserved net treatment benefit.[121] Perioperative management is also often suboptimal with excessive periods of withholding anticoagulation because of application of warfarin protocols to direct oral anticoagulants (which generally only need 24–48 h cessation preoperatively[120]) or postponement of surgery time.

Carotid endarterectomy is the preferred procedure for symptomatic carotid stenosis of 70–99% with smaller but still significant benefit in patients with 50–69% symptomatic stenosis.[122] Surgery should typically be done within 2 weeks of the index stroke or transient ischaemic attack and the benefits rapidly decrease with elapsed time. The benefits reported in endarterectomy trials might be reduced in clinical practice because of improved medical management; ongoing trials are seeking to refine risk stratification in the context of intensive medical therapy. Although an early trial suggested a benefit of endarterectomy in selected patients with asymptomatic carotid stenosis,[123] this trial did not reflect contemporary intensive medical management, which should be the cornerstone of management. An ongoing trial is assessing intervention in the setting of maximal medical management (CREST2, NCT03385928).[124]

Carotid stenting has a role in patients with unfavourable anatomy, restenosis of endarterectomy, high perioperative risk, previous radiotherapy, or other factors that would increase the risk of endarterectomy. To date, stenting in patients who are also eligible for endarterectomy has shown consistently higher risk of periprocedural stroke than endarterectomy,[125,126] with possible exception of patients aged younger than 70 years.[127] Stenting is mainly used in the context of emergency endovascular thrombectomy. However, transcarotid stenting, which involves direct percutaneous access to the common carotid (avoiding traversing the aortic arch), and flow reversal before crossing the stenosis, appeared to have lower perioperative stroke risk in initial observational data than did transfemoral carotid stenting[128] and could become a preferred approach.

Percutaneous closure of a patent foramen ovale for patients younger than 60 years with no other identified cause of stroke is now supported by the results of multiple randomised trials.[98] The coexistence of an atrial septal aneurysm (hypermobile interatrial septum) portends a higher risk of recurrent stroke than for a patent foramen ovale without this aneurysm.[129] The main risk of closure is atrial fibrillation, which occurs in approximately 2·4% of patients but is usually transient.[98] Patients are prescribed aspirin and clopidogrel for 3–6 months to reduce the risk of device thrombus pending endothelialisation.

Mechanical occlusion of the left atrial appendage might be beneficial in some patients with atrial fibrillation and a genuine contraindication to anticoagulation. Approximately 90% of thromboemboli in atrial fibrillation originate from the left atrial appendage. Randomised trials have suggested similar stroke prevention to anticoagulation.[130] In these trials, patients still required anticoagulation in the periprocedural period, although dual antiplatelet therapy with aspirin and clopidogrel has been used in practice.[131] Left atrial appendage occlusion

does, however, provide an alternative to long-term anticoagulation.

Intracerebral haemorrhage

Distinguishing the specific mechanism of intracerebral haemorrhage is increasingly recognised as clinically relevant, rather than accepting a classification as primary intracerebral haemorrhage.[20] A CT angiogram can rapidly exclude most aneurysms and arteriovenous malformations. MRI with contrast done approximately 3 months after intracerebral haemorrhage is helpful to ensure the expected evolution of haematoma and exclude an underlying mass lesion or a vascular malformation that was initially compressed by the haematoma. MRI can also show an underlying deep perforating vasculopathy or amyloid angiopathy. In the absence of evidence of angiopathy, further investigation with catheter angiography might be warranted to exclude small vascular malformations.

Lowering blood pressure is the mainstay of secondary prevention after intracerebral haemorrhage. A reduction of approximately 9 mm Hg in systolic blood pressure was associated with a 50% (95% CI 26–67) relative reduction in risk of intracerebral haemorrhage,[101] with no lower threshold for benefit identified.[132]

Although it might seem logical to avoid antithrombotics after intracerebral haemorrhage, atherosclerosis often coexists and there is a competing risk of ischaemic events. The RESTART trial[133] randomly assigned patients who had previous ischaemic heart disease or cerebrovascular disease to cease versus restart antiplatelet medications after they had an intracerebral haemorrhage. Importantly, restarting antiplatelets was associated with a non-significant reduction in recurrent intracerebral haemorrhage events (adjusted hazard ratio 0·51 [95% CI 0·25–1·03], p=0·060). Thus, a substantial increase in bleeding related to antiplatelet use appears unlikely.

There are ongoing randomised trials of restarting anticoagulation in patients with atrial fibrillation who have intracerebral haemorrhage, and also substantial risk of ischaemic stroke (eg, ASPIRE, NCT03907046). A meta-analysis of observational studies suggested that the balance of risk might favour anticoagulation overall.[134] Clearly these data could be confounded by factors that influenced physicians' decisions whether to restart anticoagulation. Importantly, there was no benefit of using an antiplatelet agent rather than anticoagulation. The risk equation can most likely be refined by separating patients with deep perforating vasculopathy from those with amyloid angiopathy, which has a higher risk of recurrent intracerebral haemorrhage.[135] Within patients with amyloid angiopathy, a large number of microbleeds and particularly cortical haemosiderosis on MRI most likely indicates a group at particularly high risk for restarting anticoagulation.[136] Importantly, percutaneous occlusion of the left atrial appendage offers an alternative to long-term anticoagulation in these patients[130] and has not been included as a comparator in studies to date. If anticoagulation is to be restarted in these patients, use of a direct oral anticoagulant might be preferable on the basis of the lower risk of cerebral bleeding versus warfarin in other contexts. Timing for starting anticoagulation after an intracerebral haemorrhage is also based on scarce data but 4–8 weeks might be reasonable.[137]

In patients with mechanical heart valves, recommencement of anticoagulation is especially crucial as the risk of ischaemic stroke is higher, but availability of data to guide the timing of restarting anticoagulation is scarce. The largest observational study suggested that the optimal balance of risks occurred with recommencement 1–2 weeks following intracerebral haemorrhage.[138]

Rehabilitation and recovery

For people who have had stroke, the ability to return to work and social functions is the key priority. Structured rehabilitation is the accepted practice in most high-income countries but is non-existent in many low-income or middle-income regions where the family are responsible for postacute care. Developing evidence for rehabilitation interventions has been challenging. Most randomised trials have not shown a benefit of the intervention of interest. For example, the largest trial for stroke rehabilitation showed a harmful effect of early intensive mobilisation within 24 h of stroke onset.[139] Constraint-induced movement therapy is one of the few rehabilitation interventions that is supported by evidence from randomised trials, improving limb function but not significantly reducing disability.[140–142] There is increasing recognition of the heterogeneity among patients with stroke and the influence of spontaneous recovery trajectories. Researchers in the field are actively investigating biomarkers to better select and stratify patients for future physical and pharmacological strategies to enhance recovery after stroke. Robotics and other approaches to increase task repetition and the daily dose of physical therapies are investigational, as are the use of stem cells and other pharmacological approaches to induce a microenvironment that promotes recovery.

Conclusions

Care for patients with stroke has transformed over the past 5 years, particularly with reperfusion therapies for ischaemic stroke and improved secondary prevention, although large gaps between evidence and practice still exist. Interventions for intracerebral haemorrhage might similarly revolutionise our approach to that condition in the future. There is reinvigorated interest in the fields of cytoprotection and recovery enhancement. Improved implementation of our existing knowledge about prevention and rapid treatment of patients with stroke could substantially reduce the major global burden of disability related to stroke.

Contributors
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References

1 Feigin VL, Nguyen G, Cercy K, et al. Global, regional, and country-specific lifetime risks of stroke, 1990 and 2016. *N Engl J Med* 2018; **379:** 2429–37.
2 Sacco RL, Kasner SE, Broderick JP, et al. An updated definition of stroke for the 21st century: a statement for healthcare professionals from the American Heart Association/American Stroke Association. *Stroke* 2013; **44:** 2064–89.
3 GBD 2016 Stroke Collaborators. Global, regional, and national burden of stroke, 1990–2016: a systematic analysis for the Global Burden of Disease Study 2016. *Lancet Neurol* 2019; **18:** 439–58.
4 Zhang LF, Yang J, Hong Z, et al. Proportion of different subtypes of stroke in China. *Stroke* 2003; **34:** 2091–96.
5 Calvet D, Touze E, Oppenheim C, Turc G, Meder JF, Mas JL. DWI lesions and TIA etiology improve the prediction of stroke after TIA. *Stroke* 2009; **40:** 187–92.
6 Arch AE, Weisman DC, Coca S, Nystrom KV, Wira CR 3rd, Schindler JL. Missed ischemic stroke diagnosis in the emergency department by emergency medicine and neurology services. *Stroke* 2016; **47:** 668–73.
7 Hand PJ, Kwan J, Lindley RI, Dennis MS, Wardlaw JM. Distinguishing between stroke and mimic at the bedside: the Brain Attack Study. *Stroke* 2006; **37:** 769–75.
8 O'Donnell MJ, Xavier D, Liu L, et al. Risk factors for ischaemic and intracerebral haemorrhagic stroke in 22 countries (the INTERSTROKE study): a case-control study. *Lancet* 2010; **376:** 112–23.
9 Pandian JD, Gall SL, Kate MP, et al. Prevention of stroke: a global perspective. *Lancet* 2018; **392:** 1269–78.
10 Chugh SS, Havmoeller R, Narayanan K, et al. Worldwide epidemiology of atrial fibrillation: a Global Burden of Disease 2010 Study. *Circulation* 2014; **129:** 837–47.
11 Tu HT, Campbell BC, Christensen S, et al. Pathophysiological determinants of worse stroke outcome in atrial fibrillation. *Cerebrovasc Dis* 2010; **30:** 389–95.
12 Kim JS, Kim YJ, Ahn SH, Kim BJ. Location of cerebral atherosclerosis: why is there a difference between East and West? *Int J Stroke* 2018; **13:** 35–46.
13 Astrup J, Siesjo BK, Symon L. Thresholds in cerebral ischemia—the ischemic penumbra. *Stroke* 1981; **12:** 723–25.
14 Warach S, Dashe JF, Edelman RR. Clinical outcome in ischemic stroke predicted by early diffusion-weighted and perfusion magnetic resonance imaging: a preliminary analysis. *J Cereb Blood Flow Metab* 1996; **16:** 53–59.
15 Campbell BCV, Christensen S, Levi CR, et al. Comparison of computed tomography perfusion and magnetic resonance imaging perfusion-diffusion mismatch in ischemic stroke. *Stroke* 2012; **43:** 2648–53.
16 Albers GW, Marks MP, Kemp S, et al. Thrombectomy for stroke at 6 to 16 hours with selection by perfusion imaging. *N Engl J Med* 2018; **378:** 708–18.
17 Campbell BCV, Ma H, Ringleb P, et al. Extending thrombolysis to 4·5–9 hours and wake-up stroke using perfusion imaging: a systematic review and meta-analysis of individual patientdata. *Lancet* 2019; **394:** 139–47.
18 Bai J, Lyden PD. Revisiting cerebral postischemic reperfusion injury: new insights in understanding reperfusion failure, hemorrhage, and edema. *Int J Stroke* 2015; **10:** 143–52.
19 George PM, Steinberg GK. Novel stroke therapeutics: unraveling stroke pathophysiology and its impact on clinical treatments. *Neuron* 2015; **87:** 297–309.
20 Cordonnier C, Demchuk A, Ziai W, Anderson CS. Intracerebral haemorrhage: current approaches to acute management. *Lancet* 2018; **392:** 1257–68.
21 Langhorne P, Kamachandra S. Organised inpatient (stroke unit) care for stroke: network meta-analysis. *Cochrane Database Syst Rev* 2020; **4:** CD000197.
22 Emberson J, Lees KR, Lyden P, et al. Effect of treatment delay, age, and stroke severity on the effects of intravenous thrombolysis with alteplase for acute ischaemic stroke: a meta-analysis of individual patient data from randomised trials. *Lancet* 2014; **384:** 1929–35.
23 Whiteley WN, Emberson J, Lees KR, et al. Risk of intracerebral haemorrhage with alteplase after acute ischaemic stroke: a secondary analysis of an individual patient data meta-analysis. *Lancet Neurol* 2016; **15:** 925–33.
24 Goyal M, Menon BK, van Zwam WH, et al. Endovascular thrombectomy after large-vessel ischaemic stroke: a meta-analysis of individual patient data from five randomised trials. *Lancet* 2016; **387:** 1723–31.
25 Snelling B, Mccarthy DJ, Chen S, et al. Extended window for stroke thrombectomy. *J Neurosci Rural Pract* 2019; **10:** 294–300.
26 Menon BK, Hill MD, Davalos A, et al. Efficacy of endovascular thrombectomy in patients with M2 segment middle cerebral artery occlusions: meta-analysis of data from the HERMES collaboration. *J Neurointerv Surg* 2019; **11:** 1065–69.
27 Chen Z, Sandercock P, Pan H, et al. Indications for early aspirin use in ischemic stroke—a combined analysis of 40 000 randomised patients from the Chinese Acute Stroke Trial and the International Stroke Trial. *Stroke* 2000; **31:** 1240–49.
28 Anderson CS, Heeley E, Huang Y, et al. Rapid blood-pressure lowering in patients with acute intracerebral hemorrhage. *N Engl J Med* 2013; **368:** 2355–65.
29 Gregson BA, Broderick JP, Auer LM, et al. Individual patient data subgroup meta-analysis of surgery for spontaneous supratentorial intracerebral hemorrhage. *Stroke* 2012; **43:** 1496–504.
30 Scaggiante J, Zhang X, Mocco J, Kellner CP. Minimally invasive surgery for intracerebral hemorrhage. *Stroke* 2018; **49:** 2612–20.
31 Wahlgren N, Ahmed N, Davalos A, et al. Thrombolysis with alteplase for acute ischaemic stroke in the Safe Implementation of Thrombolysis in Stroke-Monitoring Study (SITS-MOST): an observational study. *Lancet* 2007; **369:** 275–82.
32 Middleton S, McElduff P, Ward J, et al. Implementation of evidence-based treatment protocols to manage fever, hyperglycaemia, and swallowing dysfunction in acute stroke (QASC): a cluster randomised controlled trial. *Lancet* 2011; **378:** 1699–706.
33 Johnston KC, Bruno A, Pauls Q, et al. Intensive vs standard treatment of hyperglycemia and functional outcome in patients with acute ischemic stroke: the SHINE randomized clinical trial. *JAMA* 2019; **322:** 326–35.
34 The National Institute of Neurological Disorders and Stroke rt-PA Stroke Study Group. Tissue plasminogen activator for acute ischemic stroke. *N Engl J Med* 1995; **333:** 1581–87.
35 Lansberg MG, Schrooten M, Bluhmki E, Thijs VN, Saver JL. Treatment time-specific number needed to treat estimates for tissue plasminogen activator therapy in acute stroke based on shifts over the entire range of the modified Rankin Scale. *Stroke* 2009; **40:** 2079–84.
36 Khatri P, Kleindorfer DO, Devlin T, et al. Effect of alteplase vs aspirin on functional outcome for patients with acute ischemic stroke and minor nondisabling neurologic deficits: the PRISMS randomized clinical trial. *JAMA* 2018; **320:** 156–66.
37 Ma H, Campbell BCV, Parsons MW, et al. Thrombolysis guided by perfusion imaging up to 9 hours after onset of stroke. *N Engl J Med* 2019; **380:** 1795–803.
38 Thomalla G, Simonsen CZ, Boutitie F, et al. MRI-guided thrombolysis for stroke with unknown time of onset. *N Engl J Med* 2018; **379:** 611–22.
39 Barow E, Boutitie F, Cheng B, et al. Functional outcome of intravenous thrombolysis in patients with lacunar infarcts in the WAKE-UP trial. *JAMA Neurol* 2019; **76:** 641–49.
40 Yamaguchi T, Mori E, Minematsu K, et al. Alteplase at 0·6 mg/kg for acute ischemic stroke within 3 hours of onset: Japan alteplase clinical trial (J-ACT). *Stroke* 2006; **37:** 1810–15.
41 Anderson CS, Robinson T, Lindley RI, et al. Low-dose versus standard-dose intravenous alteplase in acute ischemic stroke. *N Engl J Med* 2016; **374:** 2313–23.
42 Tanswell P, Modi N, Combs D, Danays T. Pharmacokinetics and pharmacodynamics of tenecteplase in fibrinolytic therapy of acute myocardial infarction. *Clin Pharmacokinet* 2002; **41:** 1229–45.

43 Burgos AM, Saver JL. Evidence that tenecteplase is noninferior to alteplase for acute ischemic stroke. *Stroke* 2019; **50:** 2156–62.

44 Bivard A, Huang X, Levi CR, et al. Tenecteplase in ischemic stroke offers improved recanalization: analysis of 2 trials. *Neurology* 2017; **89:** 62–67.

45 Campbell BCV, Mitchell PJ, Churilov L, et al. Tenecteplase versus alteplase before thrombectomy for ischemic stroke. *N Engl J Med* 2018; **378:** 1573–82.

46 Campbell BCV, Mitchell PJ, Churilov L, et al. Effect of intravenous tenecteplase dose on cerebral reperfusion before thrombectomy in patients with large vessel occlusion ischemic stroke: the EXTEND-IA TNK part 2 randomized clinical trial. *JAMA* 2020; **323:** 1257–65.

47 Turc G, Bhogal P, Fischer U, et al. European Stroke Organisation (ESO)—European Society for Minimally Invasive Neurological Therapy (ESMINT) guidelines on mechanical thrombectomy in acute ischaemic stroke endorsed by Stroke Alliance for Europe (SAFE). *Eur Stroke J* 2019; **4:** 6–12.

48 Powers WJ, Rabinstein AA, Ackerson T, et al. Guidelines for the early management of patients with acute ischemic stroke: 2019 update to the 2018 guidelines for the early management of acute ischemic stroke: a guideline for healthcare professionals from the American Heart Association/American Stroke Association. *Stroke* 2019; **50:** e344–418.

49 Stroke Foundation (Australia). Clinical guidelines for stroke management. 2019. https://informme.org.au/Guidelines (accessed Dec 20, 2019).

50 Ramakrishnan TCR, Kumaravelu S, Narayan SK, et al. Efficacy and safety of intravenous tenecteplase bolus in acute ischemic stroke: results of two open-label, multicenter trials. *Am J Cardiovasc Drugs* 2018; **18:** 387–95.

51 Kliche W, Krech I, Michel MC, Sangole NV, Sathaye S. Comparison of clot lysis activity and biochemical properties of originator tenecteplase (Metalyse) with those of an alleged biosimilar. *Front Pharmacol* 2014; **5:** 7.

52 Yang P, Zhang Y, Zhang L, et al. Endovascular thrombectomy with or without intravenous alteplase in acute stroke. *N Engl J Med* 2020; **382:** 1981–93.

53 Martinod K, Wagner DD. Thrombosis: tangled up in NETs. *Blood* 2014; **123:** 2768–76.

54 Broderick JP, Palesch YY, Demchuk AM, et al. Endovascular therapy after intravenous t-PA versus t-PA alone for stroke. *N Engl J Med* 2013; **368:** 893–903.

55 Ciccone A, Valvassori L, Nichelatti M, et al. Endovascular treatment for acute ischemic stroke. *N Engl J Med* 2013; **368:** 904–13.

56 Kidwell CS, Jahan R, Gornbein J, et al. A trial of imaging selection and endovascular treatment for ischemic stroke. *N Engl J Med* 2013; **368:** 914–23.

57 Berkhemer OA, Fransen PS, Beumer D, et al. A randomized trial of intraarterial treatment for acute ischemic stroke. *N Engl J Med* 2015; **372:** 11–20.

58 Campbell BC, Mitchell PJ, Kleinig TJ, et al. Endovascular therapy for ischemic stroke with perfusion-imaging selection. *N Engl J Med* 2015; **372:** 1009–18.

59 Goyal M, Demchuk AM, Menon BK, et al. Randomized assessment of rapid endovascular treatment of ischemic stroke. *N Engl J Med* 2015; **372:** 1019–30.

60 Saver JL, Goyal M, Bonafe A, et al. Stent-retriever thrombectomy after intravenous t-PA vs t-PA alone in stroke. *N Engl J Med* 2015; **372:** 2285–95.

61 Jovin TG, Chamorro A, Cobo E, et al. Thrombectomy within 8 hours after symptom onset in ischemic stroke. *N Engl J Med* 2015; **372:** 2296–306.

62 Saver JL, Goyal M, van der Lugt A, et al. Time to treatment with endovascular thrombectomy and outcomes from ischemic stroke: a meta-analysis. *JAMA* 2016; **316:** 1279–88.

63 Nogueira RG, Jadhav AP, Haussen DC, et al. Thrombectomy 6 to 24 hours after stroke with a mismatch between deficit and infarct. *N Engl J Med* 2018; **378:** 11–21.

64 Liu X, Xu G, Liu Y, et al. Acute basilar artery occlusion: endovascular interventions versus standard medical treatment (BEST) trial-design and protocol for a randomized, controlled, multicenter study. *Int J Stroke* 2017; **12:** 779–85.

65 BASICS study group. A randomised acute stroke trial of endovascular therapy in acute basilar artery occlusion. May, 2020. https://eso-wso-conference.org/eso-wso-may-webinar (accessed June 26, 2020).

66 Campbell BCV, Majoie C, Albers GW, et al. Penumbral imaging and functional outcome in patients with anterior circulation ischaemic stroke treated with endovascular thrombectomy versus medical therapy: a meta-analysis of individual patient-level data. *Lancet Neurol* 2019; **18:** 46–55.

67 Roman LS, Menon BK, Blasco J, et al. Imaging features and safety and efficacy of endovascular stroke treatment: a meta-analysis of individual patient-level data. *Lancet Neurol* 2018; **17:** 895–904.

68 Fisher M, Feuerstein G, Howells DW, et al. Update of the stroke therapy academic industry roundtable preclinical recommendations. *Stroke* 2009; **40:** 2244–50.

69 Hill MD, Goyal M, Menon BK, et al. Efficacy and safety of nerinetide for the treatment of acute ischaemic stroke (ESCAPE-NA1): a multicentre, double-blind, randomised controlled trial. *Lancet* 2020; **395:** 878–87.

70 Cook DJ, Teves L, Tymianski M. Treatment of stroke with a PSD-95 inhibitor in the gyrencephalic primate brain. *Nature* 2012; **483:** 213–17.

71 Hill MD, Martin RH, Mikulis D, et al. Safety and efficacy of NA-1 in patients with iatrogenic stroke after endovascular aneurysm repair (ENACT): a phase 2, randomised, double-blind, placebo-controlled trial. *Lancet Neurol* 2012; **11:** 942–50.

72 Vahedi K, Hofmeijer J, Juettler E, et al. Early decompressive surgery in malignant infarction of the middle cerebral artery: a pooled analysis of three randomised controlled trials. *Lancet Neurol* 2007; **6:** 215–22.

73 Qureshi AI, Palesch YY, Barsan WG, et al. Intensive blood-pressure lowering in patients with acute cerebral hemorrhage. *N Engl J Med* 2016; **375:** 1033–43.

74 Moullaali TJ, Wang X, Martin RH, et al. Blood pressure control and clinical outcomes in acute intracerebral haemorrhage: a preplanned pooled analysis of individual participant data. *Lancet Neurol* 2019; **18:** 857–64.

75 Steiner T, Poli S, Griebe M, et al. Fresh frozen plasma versus prothrombin complex concentrate in patients with intracranial haemorrhage related to vitamin K antagonists (INCH): a randomised trial. *Lancet Neurol* 2016; **15:** 566–73.

76 Baharoglu MI, Cordonnier C, Salman RA, et al. Platelet transfusion versus standard care after acute stroke due to spontaneous cerebral haemorrhage associated with antiplatelet therapy (PATCH): a randomised, open-label, phase 3 trial. *Lancet* 2016; **387:** 2605–13.

77 Sprigg N, Flaherty K, Appleton JP, et al. Tranexamic acid for hyperacute primary intracerebral haemorrhage (TICH-2): an international randomised, placebo-controlled, phase 3 superiority trial. *Lancet* 2018; **391:** 2107–15.

78 Mayer SA, Brun NC, Begtrup K, et al. Efficacy and safety of recombinant activated factor VII for acute intracerebral hemorrhage. *N Engl J Med* 2008; **358:** 2127–37.

79 Mayer SA, Brun NC, Begtrup K, et al. Recombinant activated factor VII for acute intracerebral hemorrhage. *N Engl J Med* 2005; **352:** 777–85.

80 The CRASH-3 Trial Collaborators. Effects of tranexamic acid on death, disability, vascular occlusive events and other morbidities in patients with acute traumatic brain injury (CRASH-3): a randomised, placebo-controlled trial. *Lancet* 2019; **394:** 1713–23.

81 Hanley DF, Thompson RE, Rosenblum M, et al. Efficacy and safety of minimally invasive surgery with thrombolysis in intracerebral haemorrhage evacuation (MISTIE III): a randomised, controlled, open-label, blinded endpoint phase 3 trial. *Lancet* 2019; **393:** 1021–32.

82 Kleindorfer DO, Miller R, Moomaw CJ, et al. Designing a message for public education regarding stroke: does FAST capture enough stroke? *Stroke* 2007; **38:** 2864–68.

83 Meretoja A, Weir L, Ugalde M, et al. Helsinki model cut stroke thrombolysis delays to 25 minutes in Melbourne in only 4 months. *Neurology* 2013; **81:** 1071–76.

84 Perez de la Ossa N, Carrera D, Gorchs M, et al. Design and validation of a prehospital stroke scale to predict large arterial occlusion: the rapid arterial occlusion evaluation scale. *Stroke* 2014; **45:** 87–91.

85 Noorian AR, Sanossian N, Shkirkova K, et al. Los Angeles motor scale to identify large vessel occlusion: prehospital validation and comparison with other screens. *Stroke* 2018; **49:** 565–72.

86 Zhao H, Pesavento L, Coote S, et al. Ambulance clinical triage for acute stroke treatment: paramedic triage algorithm for large vessel occlusion. *Stroke* 2018; **49:** 945–51.

87 Panagos P, Schwamm L. Mission lifeline: severity-based stroke triage algorithm for EMS. 2020. https://www.heart.org/en/professional/quality-improvement/mission-lifeline/mission-lifeline-stroke (accessed April 7, 2020).

88 Holodinsky JK, Williamson TS, Demchuk AM, et al. Modeling stroke patient transport for all patients with suspected large-vessel occlusion. *JAMA Neurol* 2018; **75:** 1477–86.

89 Walter S, Kostopoulos P, Haass A, et al. Diagnosis and treatment of patients with stroke in a mobile stroke unit versus in hospital: a randomised controlled trial. *Lancet Neurol* 2012; **11:** 397–404.

90 Ebinger M, Winter B, Wendt M, et al. Effect of the use of ambulance-based thrombolysis on time to thrombolysis in acute ischemic stroke: a randomized clinical trial. *JAMA* 2014; **311:** 1622–31.

91 Audebert H, Ebinger M, Siegerink B, et al. The effects of mobile stroke units on functional outcome after acute cerebral ischemia. 2020. https://professional.heart.org/idc/groups/ahamah-public/@wcm/@sop/@scon/documents/downloadable/ucm_505623.pdf (accessed June 30, 2020).

92 Fonarow GC, Zhao X, Smith EE, et al. Door-to-needle times for tissue plasminogen activator administration and clinical outcomes in acute ischemic stroke before and after a quality improvement initiative. *JAMA* 2014; **311:** 1632–40.

93 Campbell BC, Weir L, Desmond PM, et al. CT perfusion improves diagnostic accuracy and confidence in acute ischaemic stroke. *J Neurol Neurosurg Psychiatry* 2013; **84:** 613–18.

94 Froehler MT, Saver JL, Zaidat OO, et al. Interhospital transfer before thrombectomy is associated with delayed treatment and worse outcome in the STRATIS registry (systematic evaluation of patients treated with neurothrombectomy devices for acute ischemic stroke). *Circulation* 2017; **136:** 2311–21.

95 Choi PMC, Tsoi AH, Pope AL, et al. Door-in-door-out time of 60 minutes for stroke with emergent large vessel occlusion at a primary stroke center. *Stroke* 2019; **50:** 2829–34.

96 Sanna T, Diener HC, Passman RS, et al. Cryptogenic stroke and underlying atrial fibrillation. *N Engl J Med* 2014; **370:** 2478–86.

97 Ziegler PD, Rogers JD, Ferreira SW, et al. Long-term detection of atrial fibrillation with insertable cardiac monitors in a real-world cryptogenic stroke population. *Int J Cardiol* 2017; **244:** 175–79.

98 Turc G, Calvet D, Guerin P, et al. Closure, anticoagulation, or antiplatelet therapy for cryptogenic stroke with patent foramen ovale: systematic review of randomized trials, sequential meta-analysis, and new insights from the CLOSE study. *J Am Heart Assoc* 2018; **7:** e008356.

99 Palazzo P, Ingrand P, Agius P, Belhadj Chaidi R, Neau JP. Transcranial doppler to detect right-to-left shunt in cryptogenic acute ischemic stroke. *Brain Behav* 2019; **9:** e01091.

100 Prospective Studies Collaboration. Age-specific relevance of usual blood pressure to vascular mortality: a meta-analysis of individual data for one million adults in 61 prospective studies. *Lancet* 2002; **360:** 1903–13.

101 PROGRESS Collaborative Group. Randomised trial of a perindopril-based blood-pressure-lowering regimen among 6105 individuals with previous stroke or transient ischaemic attack. *Lancet* 2001; **358:** 1033–41.

102 Wright JT Jr, Williamson JD, Whelton PK, et al. A randomized trial of intensive versus standard blood-pressure control. *N Engl J Med* 2015; **373:** 2103–16.

103 The SPS3 Study Group. Blood-pressure targets in patients with recent lacunar stroke: the SPS3 randomised trial. *Lancet* 2013; **382:** 507–15.

104 Sandset EC, Bath PM, Boysen G, et al. The angiotensin-receptor blocker candesartan for treatment of acute stroke (SCAST): a randomised, placebo-controlled, double-blind trial. *Lancet* 2011; **377:** 741–50.

105 Andrew NE, Kim J, Thrift AG, et al. Prescription of antihypertensive medication at discharge influences survival following stroke. *Neurology* 2018; **90:** e745–53.

106 National Institute for Health and Care Excellence. Hypertension in adults: diagnosis and management. 2019. https://www.nice.org.uk/guidance/ng136/chapter/Recommendations (accessed March 30, 2020).

107 Webb AJ, Fischer U, Mehta Z, Rothwell PM. Effects of antihypertensive-drug class on interindividual variation in blood pressure and risk of stroke: a systematic review and meta-analysis. *Lancet* 2010; **375:** 906–15.

108 Kernan WN, Ovbiagele B, Black HR, et al. Guidelines for the prevention of stroke in patients with stroke and transient ischemic attack: a guideline for healthcare professionals from the American Heart Association/American Stroke Association. *Stroke* 2014; **45:** 2160–236.

109 Amarenco P, Bogousslavsky J, Callahan A 3rd, et al. High-dose atorvastatin after stroke or transient ischemic attack. *N Engl J Med* 2006; **355:** 549–59.

110 Choi KH, Seo WK, Park MS, et al. Effect of statin therapy on outcomes of patients with acute ischemic stroke and atrial fibrillation. *J Am Heart Assoc* 2019; **8:** e013941.

111 Amarenco P, Kim JS, Labreuche J, et al. A comparison of two LDL cholesterol targets after ischemic stroke. *N Engl J Med* 2020; **382:** 9.

112 Sabatine MS, Giugliano RP, Pedersen TR. Evolocumab in patients with cardiovascular disease. *N Engl J Med* 2017; **377:** 787–88.

113 Hart RG, Sharma M, Mundl H, et al. Rivaroxaban for stroke prevention after embolic stroke of undetermined source. *N Engl J Med* 2018; **378:** 2191–201.

114 Diener HC, Sacco RL, Easton JD, et al. Dabigatran for prevention of stroke after embolic stroke of undetermined source. *N Engl J Med* 2019; **380:** 1906–17.

115 Jalini S, Rajalingam R, Nisenbaum R, Javier AD, Woo A, Pikula A. Atrial cardiopathy in patients with embolic strokes of unknown source and other stroke etiologies. *Neurology* 2019; **92:** e288–94.

116 Wang Y, Zhao X, Liu L, et al. Clopidogrel with aspirin in acute minor stroke or transient ischemic attack. *N Engl J Med* 2013; **369:** 11–19.

117 Johnston SC, Elm JJ, Easton JD, et al. Time course for benefit and risk of clopidogrel and aspirin after acute transient ischemic attack and minor ischemic stroke: a secondary analysis from the POINT randomized trial. *Circulation* 2019; **140:** 658–64.

118 CAPRIE Steering Committee. A randomised, blinded, trial of clopidogrel versus aspirin in patients at risk of ischaemic events (CAPRIE). *Lancet* 1996; **348:** 1329–39.

119 ESPRIT Steering Group, Halkes PH, van Gijn J, Kappelle LJ, Koudstaal PJ, Algra A. Aspirin plus dipyridamole versus aspirin alone after cerebral ischaemia of arterial origin (ESPRIT): randomised controlled trial. *Lancet* 2006; **367:** 1665–73.

120 Kirchhof P, Benussi S, Kotecha D, et al. 2016 ESC guidelines for the management of atrial fibrillation developed in collaboration with EACTS. *Eur Heart J* 2016; **37:** 2893–962.

121 Friberg L, Rosenqvist M, Lip GY. Net clinical benefit of warfarin in patients with atrial fibrillation: a report from the Swedish atrial fibrillation cohort study. *Circulation* 2012; **125:** 2298–307.

122 Orrapin S, Rerkasem K. Carotid endarterectomy for symptomatic carotid stenosis. *Cochrane Database Syst Rev* 2017; **6:** CD001081.

123 Halliday A, Harrison M, Hayter E, et al. 10-year stroke prevention after successful carotid endarterectomy for asymptomatic stenosis (ACST-1): a multicentre randomised trial. *Lancet* 2010; **376:** 1074–84.

124 Chambers BR, Donnan GA. Carotid endarterectomy for asymptomatic carotid stenosis. *Cochrane Database Syst Rev* 2005; **4:** CD001923.

125 International Carotid Stenting Study Investigators. Carotid artery stenting compared with endarterectomy in patients with symptomatic carotid stenosis (International Carotid Stenting Study): an interim analysis of a randomised controlled trial. *Lancet* 2010; **375:** 985–97.

126 Brott TG, Hobson RW 2nd, Howard G, et al. Stenting versus endarterectomy for treatment of carotid-artery stenosis. *N Engl J Med* 2010; **363:** 11–23.

127 Carotid Stenting Trialists' Collaboration. Short-term outcome after stenting versus endarterectomy for symptomatic carotid stenosis: a preplanned meta-analysis of individual patient data. *Lancet* 2010; **376:** 1062–73.

128 Schermerhorn ML, Liang P, Eldrup-Jorgensen J, et al. Association of transcarotid artery revascularization vs transfemoral carotid artery stenting with stroke or death among patients with carotid artery stenosis. *JAMA* 2019; **322:** 2313–22.

129 Mas JL, Arquizan C, Lamy C, et al. Recurrent cerebrovascular events associated with patent foramen ovale, atrial septal aneurysm, or both. *N Engl J Med* 2001; **345:** 1740–46.

130 Reddy VY, Doshi SK, Kar S, et al. 5-year outcomes after left atrial appendage closure: from the PREVAIL and PROTECT AF trials. *J Am Coll Cardiol* 2017; **70:** 2964–75.

131 Boersma LV, Ince H, Kische S, et al. Evaluating real-world clinical outcomes in atrial fibrillation patients receiving the WATCHMAN left atrial appendage closure technology. *Circ Arrhythm Electrophysiol* 2019; **12:** e006841.

132 Arima H, Chalmers J, Woodward M, et al. Lower target blood pressures are safe and effective for the prevention of recurrent stroke: the PROGRESS trial. *J Hypertens* 2006; **24:** 1201–08.

133 RESTART Collaboration. Effects of antiplatelet therapy after stroke due to intracerebral haemorrhage (RESTART): a randomised, open-label trial. *Lancet* 2019; **393:** 2613–23.

134 Korompoki E, Filippidis FT, Nielsen PB, et al. Long-term antithrombotic treatment in intracranial hemorrhage survivors with atrial fibrillation. *Neurology* 2017; **89:** 687–96.

135 Charidimou A, Imaizumi T, Moulin S, et al. Brain hemorrhage recurrence, small vessel disease type, and cerebral microbleeds: a meta-analysis. *Neurology* 2017; **89:** 820–29.

136 Charidimou A, Boulouis G, Roongpiboonsopit D, et al. Cortical superficial siderosis and recurrent intracerebral hemorrhage risk in cerebral amyloid angiopathy: large prospective cohort and preliminary meta-analysis. *Int J Stroke* 2019; **14:** 723–33.

137 Kuramatsu JB, Huttner HB. Management of oral anticoagulation after intracerebral hemorrhage. *Int J Stroke* 2019; **14:** 238–46.

138 Kuramatsu JB, Sembill JA, Gerner ST, et al. Management of therapeutic anticoagulation in patients with intracerebral haemorrhage and mechanical heart valves. *Eur Heart J* 2018; **39:** 1709–23.

139 The AVERT Trial Collaboration Group. Efficacy and safety of very early mobilisation within 24 h of stroke onset (AVERT): a randomised controlled trial. *Lancet* 2015; **386:** 46–55.

140 Wolf SL, Thompson PA, Winstein CJ, et al. The EXCITE stroke trial: comparing early and delayed constraint-induced movement therapy. *Stroke* 2010; **41:** 2309–15.

141 Corbetta D, Sirtori V, Castellini G, Moja L, Gatti R. Constraint-induced movement therapy for upper extremities in people with stroke. *Cochrane Database Syst Rev* 2015; **10:** CD004433.

142 Kwakkel G, Winters C, van Wegen EE, et al. Effects of unilateral upper limb training in two distinct prognostic groups early after stroke: the EXPLICIT-stroke randomized clinical trial. *Neurorehabil Neural Repair* 2016; **30:** 804–16.

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